

9(a)

any **nine** from:

- 1 (function is) to make ATP ;
- 2 *ref. to* double membrane / outer and inner membrane / envelope ;

inner membrane

- 3 folded / cristae, to increase / for large, surface area ;
- 4 has, ATP synthase / stalked particles ;
- 5 has, ETC / carrier (proteins) / cytochromes ;
- 6 (site of) oxidative phosphorylation / chemiosmosis ;
- 7 impermeable to protons ;

intermembrane space

- 8 has low pH / high concentration of protons ;
- 9 protons pumped into intermembrane space ;
- 10 proton gradient between intermembrane space and matrix
or
protons diffuse from intermembrane space to matrix ;

matrix

- 11 contains (co)enzymes for, link reaction / the Krebs cycle ;

outer membrane

- 12 permeable to, pyruvate / reduced NAD / oxygen ;
- 13 AVP ; e.g. ribosomes / DNA, involved in protein synthesis

Question	Answer	Marks
9(b)	any six from: 1 <u>aerenchyma</u> ; 2 in stem and roots ; 3 help <u>oxygen</u> to, move / diffuse, to, roots / submerged parts ; 4 shallow roots ; 5 air (film) trapped on underwater leaves / described ; 6 greater internode growth or leaves or flowers grow above water level ; 7 (growth regulated by) gibberellin / ethene ; 8 anaerobic respiration, in roots / underwater / when submerged ; A alcoholic fermentation 9 tolerant to high <u>ethanol</u> (concentration) ; 10 <i>ref. to</i> ethanol / alcohol, dehydrogenase ;	6